01 · Domain 1 — Cloud Concepts (24%)

Task statements: **1.1** Benefits of the AWS Cloud · **1.2** Design principles · **1.3** Migration benefits and strategies · **1.4** Cloud economics

> **Scope tags (official guide v1.3):** `v1.3` listed in the latest guide · `LEGACY` = older guide only / not listed now · `OUT` explicitly out of scope · `NEW` new in the v1.3 list. See file 00 for the full scope map.

1.0 Foundations you must know first

What is cloud computing?

On-demand delivery of IT resources (compute, storage, databases, networking, analytics…) over the internet with pay-as-you-go pricing. Instead of buying and maintaining physical servers, you rent capacity from AWS and release it when you no longer need it.

Three cloud service models

Model	You manage	Example on AWS
IaaS	OS, middleware, apps, data	**Amazon EC2** (+ VPC, EBS)
PaaS	Your app and data	**AWS Elastic Beanstalk**, RDS (managed DB)
SaaS	Nothing but your usage	**Amazon Connect** (cloud contact center), Amazon WorkSpaces Web-style end-user services; outside AWS: Gmail, Salesforce

> Exam rule of thumb: **more managed = less control = less effort**. EC2 (you patch the OS) → Elastic Beanstalk (AWS runs the platform) → SaaS (AWS runs everything).

Three cloud deployment models

Model	Meaning	Typical case
Cloud (public cloud)	All resources run in the cloud provider	Startups, new apps
Hybrid	Cloud + on-premises, connected	Gradual migration, data that must stay on-prem, low-latency to factory
On-premises (private cloud)	All in your own data center (with virtualization/automation)	Strict regulatory control

AWS tools for hybrid: **VPN, Direct Connect, Storage Gateway, AWS Outposts**.

1.1 Benefits of the AWS Cloud

The six advantages of cloud computing

#	Advantage	What it means	Example
1	**Trade fixed expense for variable expense**	No upfront data-center investment; pay only for what you consume	Startup launches with $0 hardware cost
2	**Benefit from massive economies of scale**	AWS aggregates usage of hundreds of thousands of customers → lower prices per unit	Prices go down over time
3	**Stop guessing capacity**	Scale up or down on demand; no over/under provisioning	Black Friday traffic spike handled automatically
4	**Increase speed and agility**	Resources in minutes; experiments cost little	Spin up 100 servers for a test, delete after
5	**Stop spending money running and maintaining data centers**	Focus on customers/business, not racking servers	Redirect IT staff to features
6	**Go global in minutes**	Deploy in multiple Regions with a few clicks	Low-latency for users in other continents

Key vocabulary — don't mix them up

Term	Definition	AWS example
Scalability	Capacity to increase (or decrease) resources to meet load. *Vertical* = bigger instance; *horizontal* = more instances	Add EC2 instances
Elasticity	Scalability that happens **automatically**, matching resources to demand in both directions (and you stop paying when you scale in)	**EC2 Auto Scaling**, DynamoDB on-demand

Term	Definition	AWS example
High availability (HA)	Workload remains accessible with minimal downtime despite failures	Run across **multiple AZs**
Fault tolerance	Continues operating **without interruption** even if components fail (stronger than HA, usually costlier)	Redundant, replicated, multi-AZ components
Agility	Speed at which you can provision, test, and change	Launch an environment in minutes with CloudFormation
Durability	Data is not lost	S3: designed for 99.999999999% (11 nines) durability
Availability	Data/system is reachable when needed	S3 Standard: 99.99% designed availability

> Classic trap: **elasticity ≠ scalability.** Elasticity = automatic and *both directions*. If a question says "automatically scale in and out to match demand" → elasticity.

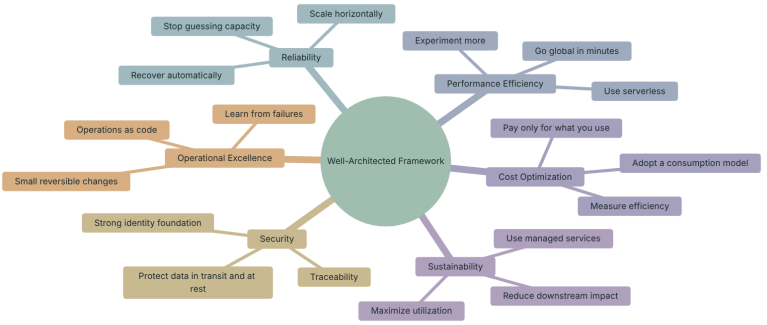
Benefits of the AWS global infrastructure

- **Speed of deployment** (infrastructure already built)
- **Global reach** (Regions worldwide, edge locations for content)
- **High availability** through multiple Availability Zones
- **Data sovereignty/compliance** — choose the Region where data lives (data does not leave a Region unless *you* move it)

1.2 Design principles — AWS Well-Architected Framework

The Well-Architected Framework is a set of **best practices** to design and operate reliable, secure, efficient, cost-effective, sustainable workloads. Tool: **AWS Well-Architected Tool** (free, in the console) — review a workload against the pillars and get an improvement plan.

The six pillars



Pillar	Core question	Key design principles	Typical services
Operational Excellence	How do I run, monitor and **continuously improve** operations?	Perform operations as code · make frequent, small, reversible changes · refine procedures · anticipate failure · learn from all operational failures	CloudFormation, CloudWatch, Systems Manager, Config
Security	How do I **protect** data, systems and assets?	Strong identity foundation (least privilege) · enable traceability · security at all layers · automate security · protect data in transit/at rest · keep people away from data · prepare for security events	IAM, KMS, GuardDuty, CloudTrail, WAF
Reliability	Can the workload **recover from failure** and meet demand?	Automatically recover from failure · test recovery procedures · scale horizontally · **stop guessing capacity** · manage change through automation	Multi-AZ, Auto Scaling, Route 53, RDS Multi-AZ, backups
Performance Efficiency	Am I using the **right resources** efficiently and adapting as needs change?	Democratize advanced technologies · go global in minutes · use serverless architectures · experiment more often · consider mechanical sympathy (pick tech that fits the goal)	Right-size EC2, CloudFront, Lambda, DynamoDB
Cost Optimization	Am I **avoiding unnecessary cost**?	Implement cloud financial management · adopt a consumption model · measure overall efficiency · stop spending on undifferentiated heavy lifting · analyze and attribute expenditure	Cost Explorer, Budgets, Savings Plans, Compute Optimizer
Sustainability	Am I **minimizing environmental impact**?	Understand your impact · establish sustainability goals · maximize utilization · adopt more efficient hardware/software · use managed services · reduce downstream impact	Right-sizing, serverless, managed services, efficient Regions

Typical question: a scenario describes an action; you must map it to a pillar.

- "Automatically recover from failure / multi-AZ" → **Reliability**
- "Encrypt data, least privilege" → **Security**
- "Delete unused resources, use Spot" → **Cost Optimization**
- "Choose right instance type, use CloudFront" → **Performance Efficiency**
- "Automate deployments, runbooks as code" → **Operational Excellence**
- "Reduce carbon footprint, higher utilization" → **Sustainability**

1.3 Migration to the AWS Cloud

Benefits of migrating (why companies do it)

Lower TCO · faster time to market · increased agility and innovation · improved security posture and compliance tooling · global reach · scalability/elasticity · reduced operational burden.

AWS Cloud Adoption Framework (AWS CAF)

A framework of **guidance and best practices** to help organizations plan and manage their cloud transformation. Organized in **six perspectives**:

- **Business-side perspectives**: Business · People · Governance
- **Technical-side perspectives**: Platform · Security · Operations

Benefits of CAF (named in the exam guide): reduced business risk · improved **ESG** (environmental, social, governance) performance · increased revenue · increased operational efficiency.

Mnemonic for perspectives: **"B-P-G-P-S-O"** → *Business People Govern Platform, Secure Operations.*

The 7 Rs of migration strategies

Strategy	Meaning	Effort	Example
Retire	Shut down apps no longer needed	None	Decommission an unused legacy system
Retain	Keep on-premises for now (revisit later)	None	App with compliance blocker or recent upgrade
Rehost ("lift and shift")	Move as-is to EC2	Low	Move VMs to EC2 using **Application Migration Service**
Relocate	Move to cloud equivalent without changes/new hardware	Low	VMware → **VMware Cloud on AWS**
Repurchase ("drop and shop")	Switch to a different product, usually SaaS	Medium	Move on-prem CRM to Salesforce
Replatform ("lift, tinker and shift")	Small optimizations, no core change	Medium	Self-managed MySQL → **Amazon RDS**; app server → Elastic Beanstalk
Refactor / Re-architect	Re-design as cloud-native	High	Monolith → microservices/serverless (Lambda, DynamoDB)

If the question says "**minimum changes / fastest**" → Rehost. "**Take advantage of cloud-native / highest benefit**" → Refactor. "**Move DB to managed service with small changes**" → Replatform. "**Replace with SaaS**" → Repurchase.

Migration tools and services

Need	Service
Discover on-prem servers, dependencies, utilization	**AWS Application Discovery Service** v1.3
Central tracking dashboard of migrations	**AWS Migration Hub**
Lift-and-shift servers to EC2	**AWS Application Migration Service (MGN)**
Migrate databases with minimal downtime	**AWS Database Migration Service (DMS)** (supports ongoing replication)
Convert schema between different DB engines (e.g., Oracle → PostgreSQL)	**AWS Schema Conversion Tool (SCT)**
Transfer very large data offline (physical device)	**AWS Snow Family** LEGACY (not on the v1.3 list — know the one-liner)
SFTP/FTPS/FTP into S3/EFS	**AWS Transfer Family** OUT (explicitly out of scope in v1.3)
Ongoing online transfer from on-prem storage	**AWS DataSync** LEGACY (not on the v1.3 list)
Dedicated network link for steady transfer	**AWS Direct Connect**
Plan with TCO	**AWS Pricing Calculator** v1.3 · **Migration Evaluator** v1.3 NEW (business case/TCO for migration)

Database replication is a recognized migration strategy: keep source and target in sync (DMS continuous replication) and cut over with minimal downtime.

1.4 Cloud economics

Fixed (CapEx) vs variable (OpEx)

	CapEx (on-prem)	OpEx (cloud)
Payment	Upfront investment	Pay as you go
Risk	Over/under-provisioned capacity	Capacity matches demand
Flexibility	Low	High
Accounting	Capital expense, depreciated	Operating expense

Costs of on-premises environments (TCO components)

Hardware purchase & refresh · data-center space · power & cooling · physical security · networking equipment · staff (admins, electricians) · software licenses · maintenance contracts · downtime/redundancy · unused capacity.

> **TCO (Total Cost of Ownership)** = all direct + indirect costs over the lifetime. AWS offers the **Pricing Calculator** to estimate AWS costs and compare to on-prem. Cloud usually wins because you stop paying for idle capacity and for undifferentiated heavy lifting.

Licensing strategies

Option	Meaning
License included	Software license cost bundled in the hourly price (e.g., Windows Server, SQL Server on EC2/RDS)
BYOL (Bring Your Own License)	Use existing licenses; often with **Dedicated Hosts** for socket/core-based licensing; **AWS License Manager** tracks usage

Rightsizing

Matching instance types and sizes to **actual** workload needs at the lowest cost. Steps: monitor utilization (CloudWatch) → get recommendations (**Compute Optimizer**, **Cost Explorer** rightsizing) → resize/terminate idle resources. Do it continuously, not once.

Automation benefits

Infrastructure as code (**AWS CloudFormation**) gives repeatable, consistent, version-controlled, reviewable deployments → fewer human errors, faster delivery, lower cost, easy to replicate environments (dev/test/prod).

Managed services

AWS operates the undifferentiated heavy lifting (patching, backups, HA, scaling) so you focus on your application.

Instead of…	Use managed…
MySQL on EC2	**Amazon RDS / Aurora**
Self-built key-value store	**Amazon DynamoDB**
Docker on self-managed hosts	**Amazon ECS / EKS** (+ **Fargate**)
Self-managed Kafka	**Amazon MSK**
Self-hosted Elasticsearch	**Amazon OpenSearch Service**

Benefits: lower operational overhead, built-in HA, faster time to market, often lower TCO. Trade-off: less low-level control.

Self-check (answer without looking)

1. Name the **six advantages of cloud computing**.
2. Difference between **scalability, elasticity, high availability, fault tolerance**?
3. List the **six Well-Architected pillars** and one design principle each.
4. List the **six CAF perspectives**.
5. Name the **7 Rs** and give an example for Rehost, Replatform, Repurchase.
6. Which service migrates a DB? Which converts the schema? Which moves petabytes offline?
7. CapEx vs OpEx — which is the cloud model?
8. BYOL vs license included — which AWS option helps with socket/core-based licensing?

Answers

1. Variable expense · economies of scale · stop guessing capacity · speed & agility · stop running data centers · go global in minutes.
2. See the table above.
3. Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, Sustainability.

4. Business, People, Governance, Platform, Security, Operations.
5. Rehost → VMs to EC2; Replatform → DB to RDS; Repurchase → CRM to SaaS.
6. DMS · SCT · Snow Family.
7. OpEx.
8. Dedicated Hosts (+ License Manager).